USN | | | | | | | | | | |

09/12/24 M/S

[22CE51]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

DESIGN OF RCC STRUCTURES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed

ii. Use of IS 456-2000 code and SP 16 charts are permitted.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define Reinforced cement concrete. What is the purpose of using reinforced cement concrete?	L2	CO1	**02**
	b.	What is meant by Lap Length and development length?	L2	CO1	**02**
	c.	What is meant by slenderness ratio of a column? Classify column based on slenderness ratio.	L2	CO2	**02**
	d.	Mention the difference between one way slab and two-way slab.	L2	CO2	**02**
	e.	Compare Long-term deflection and short-term deflection.	L2	CO1	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Describe tension zone, compression zone, effective depth, effective cover and clear cover with the help of a sketch.	L3	CO1	**05**
	b.	What are the assumptions made in the limit state of collapse in flexure?	L3	CO1	**05**
3.	**a.**	Discuss the procedure to check for the deflection and shear in beams.	L3	CO2	**05**
	b.	Describe the three cases of flanged beam analysis based on the position of Neutral Axis.	L3	CO2	**05**
4.	**a.**	Determine moment of resistance (MR) of a rectangular section reinforced with steel of area 600mm² on the tension side. The width of the beam is 200mm, effective depth is 600mm. The grade of concrete is M20 & Fe250 grade steel is used.	L3	CO3	**05**
		OR			
	b.	Determine the MR of a rectangular section of dimension 230mm X 300mm with a effective cover of 35mm to tension reinforcement. The tension reinforcement consists of 3 bars of 20mm dia bars. Assume M20 grade concrete & Fe 415 steel.	L3	CO3	**05**

Jain Deemed-to-be University Learning Resource Centre Jain Global Campus Jakkasandra-112

5.	**a.**	Determine slenderness ratio and minimum eccentricity for a column of size 450 mm x 600 mm subjected to factored load of 3000 kN. The column has an unsupported length of 3.5 m and effectively held in position and restrained against rotation in both ends (Both ends Fixed). Use M 25 concrete and Fe 415 steel.	L4	CO4	**05**
		OR			
	b.	Discuss the situations which requires adoption of doubly reinforced beam	L3	CO4	**05**
6.	**a.**	Sketch and label the reinforcement detailing of columns having lateral ties and helical reinforcement.	L3	CO5	**05**
		OR			
	b.	Sketch and label the reinforcement detailing of singly and doubly reinforced beams.	L3	CO5	**05**
7.	**a.**	Explain the necessity and importance of shear and torsional reinforcement in structural members	L3	CO6	**05**
		OR			
	b.	Explain one-way shear and two-way shear in footings with sketches.	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Derive the expression for the following parameters (with sketch) **i.** Depth of Neutral Axis **ii.** Compressive force **iii.** Tensile force **iv.** Lever Arm	L3	CO1	**10**
		OR			
	b.	Explain the various loads and forces considered while designing a structure.	L3	CO1	**10**
9.	**a.**	Explain the step-by-step design procedure of one-way slab as per IS-456: 2000.	L3	CO2	**10**
		OR			
	b.	Explain the step-by-step design procedure of doubly reinforced beam as per IS-456: 2000.	L3	CO2	**10**
10.	**a.**	Determine the ultimate moment capacity of rectangular beam having width of 280mm and overall depth 550mm. The tension and compression steel provided at an effective cover of 50mm is 2450sq.mm and 400 sq.mm respectively. Use M30 concrete and Fe500	L4	CO3	**10**

		grade steel			
		OR			
	b.	A RC beam having 250mm breadth & 500mm effective depth is provided with 3 nos. of 20mm dia bars on the tension side, assuming M20 concrete & Fe 415 steel, calculate the following: (i) N-A depth (ii) compressive force (iii) Tensile force (iv) ultimate moment (v) safe concentrated load at mid span over an effective span of 6m.	L4	CO3	**10**
11.	**a.**	Design a short rectangular tied column of width 300 mm having the maximum amount of longitudinal reinforcement, to carry an axial load of 2000 kN under service dead load and live load using M 25 and Fe 415. The column is effectively held in position at both ends and restrained against rotation at one end.	L4	CO4	**10**
		OR			
	b.	Determine the reinforcements required for a doubly reinforced beam having width 300mm and overall depth 600mm. Given factored moment 320kNm.Take effective cover as 50mm. Use M20 concrete and Fe 415 grade steel	L4	CO4	**10**
12.	**a.**	Determine the shear reinforcement and its spacing for the doubly reinforced beam having width 300mm and overall depth 600mm. Given factored moment 320kNm.Take effective cover as 50mm. Use M20 concrete and Fe 415 grade steel Check for deflection and sketch the reinforcement detailing.	L4	CO5	**10**
		OR			
	b.	Determine the unsupported length of the short column of width 300 mm having the maximum amount of longitudinal reinforcement, to carry an axial load of 2000 kN under service dead load and live load using M 25 and Fe 415. The column is effectively held in position at both ends and restrained against rotation at one end. sketch the reinforcement detailing.	L4	CO5	**10**
13.	**a.**	What do you understand by Bond Strength in RCC. Explain the importance of development length with neat sketch.	L4	CO6	**10**
		OR			
	b.	A hall has a clear dimensions 3m X 9m with wall thickness 230mm.The live load on the slab is 3 kN/m^2 and a finishing load of 1 kN/m^2 can be assumed. Use M20 Grade concrete and Fe 415 grade steel. Design the slab.	L4	CO6	**10**